
DWH


TM
expedia group™

Expedia Data Warehouse Design

Grow Data Skills

Business context (Expedia-style)

- Users search trips (destination, dates) → view results → click properties/itineraries → book hotels/flights/cars/packages → pay → may modify/cancel → loyalty accrual → post-stay reviews.
- Revenue = take rate on merchant/agency models + fees + ancillaries; net after refunds/cancellations/chargebacks.

Expedia Data Warehouse Design

Grow Data Skills

Tables & Relationships (Dimension Tables)

- **dim_date**(date_key {PK}, cal_date, year, quarter, quarter_name, month, month_name, week_of_year, day_of_month, day_of_week, day_name, is_weekend, is_month_start, is_month_end, is_quarter_start, is_quarter_end, is_year_start, is_year_end) —-> 1 row per calendar day
- **dim_time**(time_key {PK}, hour, minute, second, am_pm, hour_24, day_part) —--> 1 row per minute (or per second)
- **dim_geo**(geo_sk {PK}, country_code, country_name, region_name, state_province, city, market_id, tz_name, latitude, longitude) —-> 1 row per city (adapt as needed)
- **dim_device**(device_sk {PK}, device_type, os_name, os_version, app_web, app_version, browser_name, browser_version)
- **dim_channel**(channel_sk {PK}, source, medium, campaign, ad_group, ad_id, utm_term, utm_content, partner_id) —-> Marketing related data
- **dim_customer**(customer_sk {PK}, customer_nk, email_hash, phone_hash, first_name, last_name, country_code, loyalty_tier, preferred_currency, preferred_lang, marketing_opt_in, created_at, effective_from, effective_to, is_current) —-> SCD2 Type Dim

Expedia Data Warehouse Design

Grow Data Skills

Tables & Relationships (Dimension Tables)

- **dim_property**(property_sk {PK}, property_id_nk, property_name, brand, chain_flag, star_rating, geo_sk {FK→dim_geo.geo_sk}, capacity_rooms, checkin_time, checkout_time, amenities_bitmap, effective_from, effective_to, is_current) —-> SCD2 Type Dim
- **dim_room**(room_sk {PK}, room_type_code, bed_type, max_occupancy, smoking_allowed, view_type)
- **dim_flight**(flight_sk {PK}, carrier_code, carrier_name, route_code, cabin, refundable_flag, bag_policy_code)
- **dim_car**(car_sk {PK}, supplier, car_class, transmission, fuel_type, seats)
- **dim_promo**(promo_sk {PK}, promo_code, discount_type, discount_value, start_date_key {FK→dim_date.date_key}, end_date_key {FK→dim_date.date_key}, eligibility_rules)
- **dim_payment_method**(payment_sk {PK}, payment_method, card_network, issuer_country, three_ds_enabled)
- **dim_currency_rate**(currency_rate_sk {PK}, date_key {FK→dim_date.date_key}, currency_code, fx_to_usd, fx_to_eur, fx_to_inr, fx_source, as_of_ts) —-> 1 row per (date, currency) for reporting conversion

Expedia Data Warehouse Design

Grow Data Skills

Tables & Relationships (Fact Tables)

- **fact_search_session**(search_session_id_nk {PK}, date_key {FK→dim_date.date_key}, time_key {FK→dim_time.time_key}, customer_sk {FK→dim_customer.customer_sk}, device_sk {FK→dim_device.device_sk}, geo_sk {FK→dim_geo.geo_sk}, channel_sk {FK→dim_channel.channel_sk}, search_type, origin_city, destination_city, checkin_date_key {FK→dim_date.date_key}, checkout_date_key {FK→dim_date.date_key}, los_nights, pax_adults, pax_children, result_count, search_count, created_ts) —> 1 Row per search session

- **fact_impression**(impression_id_nk {PK}, date_key {FK→dim_date.date_key}, time_key {FK→dim_time.time_key}, search_session_id_nk {FK→fact_search_session.search_session_id_nk}, channel_sk {FK→dim_channel.channel_sk}, product_type, property_sk {FK→dim_property.property_sk}, flight_sk {FK→dim_flight.flight_sk}, car_sk {FK→dim_car.car_sk}, position, price_shown, currency_code, impressions, created_ts) —> 1 row per listing shown (per session)

- **fact_click**(click_id_nk {PK}, date_key {FK→dim_date.date_key}, time_key {FK→dim_time.time_key}, search_session_id_nk {FK→fact_search_session.search_session_id_nk}, channel_sk {FK→dim_channel.channel_sk}, product_type, property_sk {FK→dim_property.property_sk}, flight_sk {FK→dim_flight.flight_sk}, car_sk {FK→dim_car.car_sk}, price_shown, currency_code, clicks, created_ts) —-> 1 Row per listing click

Expedia Data Warehouse Design

Tables & Relationships (Fact Tables)

Grow Data Skills

- **fact_booking_item**(booking_item_id_nk {PK}, booking_id_nk, date_key_booked {FK→dim_date.date_key}, time_key_booked {FK→dim_time.time_key}, customer_sk {FK→dim_customer.customer_sk}, device_sk {FK→dim_device.device_sk}, channel_sk {FK→dim_channel.channel_sk}, geo_sk {FK→dim_geo.geo_sk}, product_type, property_sk {FK→dim_property.property_sk}, room_sk {FK→dim_room.room_sk}, flight_sk {FK→dim_flight.flight_sk}, car_sk {FK→dim_car.car_sk}, payment_sk {FK→dim_payment_method.payment_sk}, promo_sk {FK→dim_promo.promo_sk}, currency_code, reporting_currency, fx_rate_used, adult_count, child_count, rooms, nights, seats, rental_days, gbv_amount, discount_amount, tax_amount, fee_amount, ancillary_amount, take_rate_pct, net_revenue_amount, is_refundable, merchant_model, booked_ts, start_date_key {FK→dim_date.date_key}, end_date_key {FK→dim_date.date_key}, status_code, raw_promo_list) —---> 1 row per booked item (room night bundle, flight leg bundle, or car rental item)

- **fact_cancellation**(cancellation_id_nk {PK}, booking_item_id_nk {FK→fact_booking_item.booking_item_id_nk}, date_key_cancel {FK→dim_date.date_key}, time_key_cancel {FK→dim_time.time_key}, channel_sk {FK→dim_channel.channel_sk}, reason_code, cancel_fee_amount, recoverable_amount, refundable_flag, cancelled_ts) —> 1 row per cancellation event (item-level)

- **fact_payment**(payment_txn_id_nk {PK}, booking_id_nk, date_key {FK→dim_date.date_key}, time_key {FK→dim_time.time_key}, payment_sk {FK→dim_payment_method.payment_sk}, currency_code, amount_authorized, amount_captured, amount_refunded, interchange_cost, three_ds_result, auth_code, txn_type, payment_ts) —> 1 row per payment transaction (auth/capture/refund as separate rows or enum)

- **fact_refund**(refund_id_nk {PK}, booking_item_id_nk {FK→fact_booking_item.booking_item_id_nk}, date_key_refund {FK→dim_date.date_key}, time_key_refund {FK→dim_time.time_key}, currency_code, refund_amount, chargeback_flag, reason_code, refund_ts) —---> 1 row per refund instruction/settlement (if you prefer separate from payments)

Expedia Data Warehouse Design

Grow Data Skills

Business Metrices & Respective SQL Queries

1) Search-to-Booking Conversion Rate

- **Definition**: The percentage of unique search sessions that resulted in at least one booking within the reporting period.
- **Formula**: (Distinct booking IDs) ÷ (Distinct search sessions) × 100.

```
WITH searches AS (
  SELECT COUNT(DISTINCT fs.search_session_id_nk) AS search_sessions
  FROM fact_search_session fs
  JOIN dim_date d ON fs.date_key = d.date_key
  WHERE d.cal_date BETWEEN :start_date AND :end_date
),
bookings AS (
  SELECT COUNT(DISTINCT bi.booking_id_nk) AS bookings
  FROM fact_booking_item bi
  JOIN dim_date d ON bi.date_key_booked = d.date_key
  WHERE d.cal_date BETWEEN :start_date AND :end_date
)
SELECT
  s.search_sessions,
  b.bookings,
  ROUND(b.bookings * 100.0 / NULLIF(s.search_sessions, 0), 2) AS conversion_rate_percent
FROM searches s CROSS JOIN bookings b;
```

Expedia Data Warehouse Design

Grow Data Skills

Business Metrices & Respective SQL Queries

2) Click-Through Rate (Listings)

- **Definition**: The percentage of listing impressions that received a click.
- **Formula**: Total clicks ÷ Total impressions × 100.

```
WITH imps AS (
  SELECT COUNT(*) AS impressions
  FROM fact_impression fi
  JOIN dim_date d ON fi.date_key = d.date_key
  WHERE d.cal_date BETWEEN :start_date AND :end_date
),
clks AS (
  SELECT COUNT(*) AS clicks
  FROM fact_click fc
  JOIN dim_date d ON fc.date_key = d.date_key
  WHERE d.cal_date BETWEEN :start_date AND :end_date
)
SELECT
  i.impressions,
  c.clicks,
  ROUND(c.clicks * 100.0 / NULLIF(i.impressions, 0), 2) AS click_through_rate_percent
FROM imps i CROSS JOIN clks c;
```

Expedia Data Warehouse Design

Grow Data Skills

Business Metrices & Respective SQL Queries

3) Gross Booking Value (GBV)

- **Definition**: The total pre-discount, pre-refund value of all booked items in the period.
- **Formula**: Sum of gbv_amount.

```
SELECT
  SUM(bi.gbv_amount) AS gross_booking_value
FROM fact_booking_item bi
JOIN dim_date d ON bi.date_key_booked = d.date_key
WHERE d.cal_date BETWEEN :start_date AND :end_date;
```

Expedia Data Warehouse Design

Grow Data Skills

4) Net Revenue

- **Definition**: Commercial revenue recognized from bookings in the period, including take rate and fees and ancillaries, minus refunds.
- **Formula**: Σ(GBV × take_rate_pct + fee_amount + ancillary_amount) − Σ(refund_amount).

```
SELECT
  SUM(bi.gbv_amount * bi.take_rate_pct + bi.fee_amount + bi.ancillary_amount)
  - COALESCE(SUM(r.refund_amount), 0) AS net_revenue
FROM fact_booking_item bi
LEFT JOIN fact_refund r
  ON r.booking_item_id_nk = bi.booking_item_id_nk
JOIN dim_date d
  ON bi.date_key_booked = d.date_key
WHERE d.cal_date BETWEEN :start_date AND :end_date;
```

Expedia Data Warehouse Design

Grow Data Skills

5) Average Selling Price (ASP)

- **Definition**: Average item price before discounts, across all booked items in the period.
- **Formula**: GBV ÷ Number of booked items.

```
SELECT
  COUNT(*) AS booking_items,
  SUM(bi.gbv_amount) AS gbv,
  ROUND(SUM(bi.gbv_amount) / NULLIF(COUNT(*), 0), 2) AS average_selling_price
FROM fact_booking_item bi
JOIN dim_date d ON bi.date_key_booked = d.date_key
WHERE d.cal_date BETWEEN :start_date AND :end_date;
```

Expedia Data Warehouse Design

Grow Data Skills

6) Cancellation Rate

- **Definition**: Percentage of booked items that were cancelled in the period.
- **Formula**: Cancelled items ÷ Booked items × 100.

```
WITH booked AS (
  SELECT COUNT(*) AS booked_items
  FROM fact_booking_item bi
  JOIN dim_date d ON bi.date_key_booked = d.date_key
  WHERE d.cal_date BETWEEN :start_date AND :end_date
),
cancelled AS (
  SELECT COUNT(DISTINCT fc.booking_item_id_nk) AS cancelled_items
  FROM fact_cancellation fc
  JOIN dim_date d ON fc.date_key_cancel = d.date_key
  WHERE d.cal_date BETWEEN :start_date AND :end_date
)
SELECT
  b.booked_items,
  c.cancelled_items,
  ROUND(c.cancelled_items * 100.0 / NULLIF(b.booked_items, 0), 2) AS cancellation_rate_percent
FROM booked b CROSS JOIN cancelled c;
```

Expedia Data Warehouse Design

Grow Data Skills

7) Refund Rate

- **Definition**: Percentage of booked items that had any refund recorded in the period.
- **Formula**: Refunded items ÷ Booked items × 100.

```
WITH booked AS (
  SELECT COUNT(DISTINCT bi.booking_item_id_nk) AS booked_items
  FROM fact_booking_item bi
  JOIN dim_date d ON bi.date_key_booked = d.date_key
  WHERE d.cal_date BETWEEN :start_date AND :end_date
),
ref_items AS (
  SELECT COUNT(DISTINCT r.booking_item_id_nk) AS refunded_items
  FROM fact_refund r
  JOIN dim_date d ON r.date_key_refund = d.date_key
  WHERE d.cal_date BETWEEN :start_date AND :end_date
)
SELECT
  b.booked_items,
  r.refunded_items,
  ROUND(r.refunded_items * 100.0 / NULLIF(b.booked_items, 0), 2) AS refund_rate_percent
FROM booked b CROSS JOIN ref_items r;
```

Expedia Data Warehouse Design

Grow Data Skills

8) Ancillary Attachment Rate

- **Definition**: Percentage of booking items that include any ancillary amount (e.g., baggage or breakfast).
- **Formula**: Items with ancillary_amount > 0 ÷ Total items × 100.

```
SELECT
  COUNT(*) AS booking_items,
  SUM(CASE WHEN bi.ancillary_amount > 0 THEN 1 ELSE 0 END) AS items_with_ancillary,
  ROUND(SUM(CASE WHEN bi.ancillary_amount > 0 THEN 1 ELSE 0 END) * 100.0
        / NULLIF(COUNT(*), 0), 2) AS ancillary_attachment_rate_percent,
  SUM(bi.ancillary_amount) AS ancillary_revenue
FROM fact_booking_item bi
JOIN dim_date d ON bi.date_key_booked = d.date_key
WHERE d.cal_date BETWEEN :start_date AND :end_date;
```

Expedia Data Warehouse Design

Grow Data Skills

9) Take Rate Percentage

- **Definition**: Commercial take percentage realized relative to GBV, after subtracting refunds.
- **Formula**: Net Revenue ÷ GBV × 100. (Net Revenue as defined above.)

```
WITH base AS (
  SELECT
    SUM(bi.gbv_amount) AS gbv,
    SUM(bi.gbv_amount * bi.take_rate_pct + bi.fee_amount + bi.ancillary_amount)
      - COALESCE(SUM(r.refund_amount), 0) AS net_revenue
  FROM fact_booking_item bi
  LEFT JOIN fact_refund r
    ON r.booking_item_id_nk = bi.booking_item_id_nk
  JOIN dim_date d ON bi.date_key_booked = d.date_key
  WHERE d.cal_date BETWEEN :start_date AND :end_date
)
SELECT
  gbv,
  net_revenue,
  ROUND(net_revenue * 100.0 / NULLIF(gbv, 0), 2) AS take_rate_percent
FROM base;
```

Expedia Data Warehouse Design

Grow Data Skills

10) Repeat Customer Rate (12-month lookback)

Definition: The percentage of customers with at least two distinct bookings in the last 12 months.
Formula: Customers with bookings ≥ 2 ÷ Customers with ≥ 1 booking × 100.

```
WITH cust_trips AS (
  SELECT
    bi.customer_sk,
    COUNT(DISTINCT bi.booking_id_nk) AS bookings_12m
  FROM fact_booking_item bi
  JOIN dim_date d ON bi.date_key_booked = d.date_key
  WHERE d.cal_date BETWEEN date_sub(current_date(), 365) AND current_date()
  GROUP BY bi.customer_sk
)
SELECT
  COUNT(*) AS customers_with_bookings,
  SUM(CASE WHEN bookings_12m >= 2 THEN 1 ELSE 0 END) AS repeat_customers,
  ROUND(SUM(CASE WHEN bookings_12m >= 2 THEN 1 ELSE 0 END) * 100.0
        / NULLIF(COUNT(*), 0), 2) AS repeat_customer_rate_percent
FROM cust_trips;
```

Expedia Data Warehouse Design

Grow Data Skills

11) Channel Performance (Net Revenue by Marketing Channel)

- **Definition**: Net revenue contribution by marketing source/medium/campaign for the period.
- **Formula**: As in Net Revenue, grouped by channel attributes.

```
SELECT
  ch.source,
  ch.medium,
  ch.campaign,
  SUM(bi.gbv_amount) AS gross_booking_value,
  (SUM(bi.gbv_amount * bi.take_rate_pct + bi.fee_amount + bi.ancillary_amount)
   - COALESCE(SUM(r.refund_amount), 0)) AS net_revenue,
  ROUND(
    (SUM(bi.gbv_amount * bi.take_rate_pct + bi.fee_amount + bi.ancillary_amount)
     - COALESCE(SUM(r.refund_amount), 0)) * 100.0
    / NULLIF(SUM(bi.gbv_amount), 0), 2
  ) AS take_rate_percent
FROM fact_booking_item bi
JOIN dim_channel ch ON bi.channel_sk = ch.channel_sk
LEFT JOIN fact_refund r ON r.booking_item_id_nk = bi.booking_item_id_nk
JOIN dim_date d ON bi.date_key_booked = d.date_key
WHERE d.cal_date BETWEEN :start_date AND :end_date
GROUP BY ch.source, ch.medium, ch.campaign
ORDER BY net_revenue DESC;
```


SWIGGY

Swiggy Data Warehouse Design

Grow Data Skills

Business context (Expedia-style)

- Customers browse restaurants, view menus, add items to cart, place orders, pay, restaurants accept & prepare, riders get assigned, pick up, and deliver. Revenue is primarily take rate on item totals plus platform fees, minus refunds/adjustments, with strong focus on on-time delivery, cancellations, and repeat behavior.

Swiggy Data Warehouse Design

Grow Data Skills

Tables & Relationships (Dimension Tables)

- **dim_date**(date_key {PK}, cal_date, year, quarter, quarter_name, month, month_name, week_of_year, day_of_month, day_of_week, day_name, is_weekend, is_month_start, is_month_end, is_quarter_start, is_quarter_end, is_year_start, is_year_end)
- **dim_time**(time_key {PK}, hour, minute, second, am_pm, hour_24, day_part)
- **dim_geo**(geo_sk {PK}, country_code, country_name, region_name, state_province, city, market_id, tz_name, latitude, longitude)
- **dim_customer**(customer_sk {PK}, customer_nk, first_name, last_name, email_hash, phone_hash, loyalty_tier, preferred_currency, preferred_lang, marketing_opt_in, segment, risk_flag, created_at, effective_from, effective_to, is_current) —> SCD2 Type
- **dim_device**(device_sk {PK}, device_type, os_name, os_version, app_web, app_version, browser_name, browser_version)
- **dim_channel**(channel_sk {PK}, source, medium, campaign, ad_group, ad_id, utm_term, utm_content, partner_id)
- **dim_restaurant** (restaurant_sk {PK}, restaurant_id_nk, restaurant_name, brand, cuisine_primary, rating, geo_sk {FK→dim_geo.geo_sk}, commission_band, is_chain_flag, open_time, close_time, effective_from, effective_to, is_current) —-> SCD2 Type

Swiggy Data Warehouse Design

Grow Data Skills

Tables & Relationships (Dimension Tables)

- **dim_menu_item**(menu_item_sk {PK}, item_id_nk, restaurant_sk {FK→dim_restaurant.restaurant_sk}, item_name, category, veg_flag, spiciness_level, base_price, calories, allergens, effective_from, effective_to, is_current) —--> SCD2 Type

- **dim_delivery_partner**(rider_sk {PK}, rider_id_nk, full_name, phone_hash, vehicle_type, license_number, geo_sk {FK→dim_geo.geo_sk}, joining_date, effective_from, effective_to, is_current) —-> SCD2 Type

- **dim_payment_method**(payment_sk {PK}, payment_method, card_network, wallet_type, issuer_country, three_ds_enabled)

- **dim_promo**(promo_sk {PK}, promo_code, discount_type, discount_value, start_date_key {FK→dim_date.date_key}, end_date_key {FK→dim_date.date_key}, targeting_rules)

- **dim_fee_policy**(fee_policy_sk {PK}, policy_name, calculation_type, parameter_json, valid_from, valid_to)

- **dim_currency_rate**(currency_rate_sk {PK}, date_key {FK→dim_date.date_key}, currency_code, fx_to_report, fx_source, as_of_ts)

Swiggy Data Warehouse Design

Grow Data Skills

Tables & Relationships (Fact Tables)

- **fact_app_session**(session_id {PK}, date_key {FK→dim_date.date_key}, time_key {FK→dim_time.time_key}, customer_sk {FK→dim_customer.customer_sk}, device_sk {FK→dim_device.device_sk}, geo_sk {FK→dim_geo.geo_sk}, channel_sk {FK→dim_channel.channel_sk}, sessions, page_views, add_to_cart_events, checkout_events, created_ts)

- **fact_menu_impression**(impression_id {PK}, date_key {FK→dim_date.date_key}, time_key {FK→dim_time.time_key}, session_id {FK→fact_app_session.session_id}, channel_sk {FK→dim_channel.channel_sk}, restaurant_sk {FK→dim_restaurant.restaurant_sk}, menu_item_sk {FK→dim_menu_item.menu_item_sk}, impressions, price_shown, created_ts)

- **fact_add_to_cart**(add_event_id {PK}, date_key {FK→dim_date.date_key}, time_key {FK→dim_time.time_key}, session_id {FK→fact_app_session.session_id}, customer_sk {FK→dim_customer.customer_sk}, restaurant_sk {FK→dim_restaurant.restaurant_sk}, menu_item_sk {FK→dim_menu_item.menu_item_sk}, qty, item_price_atc, created_ts)

Swiggy Data Warehouse Design

Grow Data Skills

Tables & Relationships (Fact Tables)

- **fact_order_header**(order_id {PK}, date_key_ordered {FK→dim_date.date_key}, time_key_ordered {FK→dim_time.time_key}, customer_sk {FK→dim_customer.customer_sk}, device_sk {FK→dim_device.device_sk}, channel_sk {FK→dim_channel.channel_sk}, restaurant_sk {FK→dim_restaurant.restaurant_sk}, geo_sk {FK→dim_geo.geo_sk}, payment_sk {FK→dim_payment_method.payment_sk}, fee_policy_sk {FK→dim_fee_policy.fee_policy_sk}, promo_sk {FK→dim_promo.promo_sk}, gmv_amount, item_discount, restaurant_discount, platform_fee, delivery_fee, taxes, promised_dropoff_ts, actual_dropoff_ts, promised_ready_ts, ready_ts, take_rate_pct, status_code, created_ts)

- **fact_order_item**(order_id {FK→fact_order_header.order_id}, item_seq {PK}, menu_item_sk {FK→dim_menu_item.menu_item_sk}, qty, line_price, line_discount, addons_amount, veg_flag, category)

- **fact_assignment**(assignment_id {PK}, order_id {FK→fact_order_header.order_id}, restaurant_sk {FK→dim_restaurant.restaurant_sk}, offered_flag, accepted_flag, accept_ts)

- **fact_delivery_trip**(trip_id {PK}, order_id {FK→fact_order_header.order_id}, rider_sk {FK→dim_delivery_partner.rider_sk}, assigned_ts, to_pickup_start_ts, at_pickup_ts, to_dropoff_start_ts, dropped_ts, distance_km, duration_sec, surge_multiplier)

- **fact_cancellation**(cancellation_id {PK}, order_id {FK→fact_order_header.order_id}, date_key_cancel {FK→dim_date.date_key}, time_key_cancel {FK→dim_time.time_key}, cancel_party, reason_code, cancel_fee_amount, cancelled_ts)

Swiggy Data Warehouse Design

Grow Data Skills

Tables & Relationships (Fact Tables)

- **fact_payment**(payment_txn_id {PK}, order_id {FK→fact_order_header.order_id}, date_key {FK→dim_date.date_key}, time_key {FK→dim_time.time_key}, payment_sk {FK→dim_payment_method.payment_sk}, amount_authorized, amount_captured, amount_refunded, txn_type, payment_ts)

- **fact_refund**(refund_id {PK}, order_id {FK→fact_order_header.order_id}, date_key_refund {FK→dim_date.date_key}, time_key_refund {FK→dim_time.time_key}, refund_amount, chargeback_flag, reason_code, refund_ts)

- **fact_restaurant_availability**(availability_id {PK}, date_key {FK→dim_date.date_key}, restaurant_sk {FK→dim_restaurant.restaurant_sk}, open_minutes, outage_minutes, uptime_pct)

- **fact_rider_shift**(shift_id {PK}, rider_sk {FK→dim_delivery_partner.rider_sk}, date_key {FK→dim_date.date_key}, shift_duration_sec, active_duration_sec, idle_duration_sec)

Swiggy Data Warehouse Design

Grow Data Skills

Business Metrices & Respective SQL Queries

1) App-to-Order Conversion Rate -> % of unique sessions that result in ≥1 completed order in the period.

Formula: distinct orders ÷ distinct sessions × 100

```
WITH s AS (
  SELECT COUNT(DISTINCT session_id) AS sessions
  FROM fact_app_session fs
  JOIN dim_date d ON fs.date_key = d.date_key
  WHERE d.cal_date BETWEEN :start_date AND :end_date
),
o AS (
  SELECT COUNT(DISTINCT oh.order_id_nk) AS orders
  FROM fact_order_header oh
  JOIN dim_date d ON oh.date_key_ordered = d.date_key
  WHERE d.cal_date BETWEEN :start_date AND :end_date
    AND oh.status IN ('DELIVERED','PLACED','ACCEPTED')  -- placed or better
)
SELECT
  s.sessions,
  o.orders,
  ROUND(o.orders * 100.0 / NULLIF(s.sessions,0), 2) AS app_to_order_conversion_percent
FROM s CROSS JOIN o;
```

Swiggy Data Warehouse Design

Grow Data Skills

Business Metrices & Respective SQL Queries

2) Average Order Value (AOV) -> Average pre-discount basket size for completed orders.

Formula: Σ item_subtotal ÷ #orders.

```
SELECT
  COUNT(*) AS orders,
  SUM(oh.gmv_amount) AS total_gmv,
  ROUND(SUM(oh.gmv_amount) / NULLIF(COUNT(*),0), 2) AS average_order_value
FROM fact_order_header oh
JOIN dim_date d ON oh.date_key_ordered = d.date_key
WHERE d.cal_date BETWEEN :start_date AND :end_date
  AND oh.status = 'DELIVERED';
```

Swiggy Data Warehouse Design

Grow Data Skills

Business Metrices & Respective SQL Queries

3.) Platform Take Rate % -> Share of commercial revenue captured relative to GMV.

Formula: (Σ (item_subtotal × take_rate) + Σ platform_fees − Σ refunds) ÷ Σ GMV × 100.

```
WITH base AS (
  SELECT
    SUM(oh.gmv_amount) AS gmv,
    SUM(oh.gmv_amount * oh.take_rate_pct + oh.platform_fee + oh.delivery_fee)
      - COALESCE(SUM(rf.refund_amount), 0) AS net_revenue
  FROM fact_order_header oh
  LEFT JOIN fact_refund rf ON rf.order_id_nk = oh.order_id_nk
  JOIN dim_date d ON oh.date_key_ordered = d.date_key
  WHERE d.cal_date BETWEEN :start_date AND :end_date
    AND oh.status IN ('DELIVERED')
)
SELECT
  gmv,
  net_revenue,
  ROUND(net_revenue * 100.0 / NULLIF(gmv,0), 2) AS platform_take_rate_percent
FROM base;
```

Swiggy Data Warehouse Design

Grow Data Skills

Business Metrices & Respective SQL Queries

4.) On-Time Delivery Rate (OTD) -> % of delivered orders where actual_dropoff_ts ≤ promised_dropoff_ts.

Formula: on_time_deliveries ÷ delivered_orders × 100.

```
SELECT
  COUNT(*) AS delivered_orders,
  SUM(CASE WHEN oh.actual_dropoff_ts <= oh.promised_dropoff_ts THEN 1 ELSE 0 END) AS on_time_orders,
  ROUND(SUM(CASE WHEN oh.actual_dropoff_ts <= oh.promised_dropoff_ts THEN 1 ELSE 0 END) * 100.0
        / NULLIF(COUNT(*),0), 2) AS on_time_delivery_percent
FROM fact_order_header oh
JOIN dim_date d ON oh.date_key_ordered = d.date_key
WHERE d.cal_date BETWEEN :start_date AND :end_date
  AND oh.status = 'DELIVERED';
```

Swiggy Data Warehouse Design

Grow Data Skills

Business Metrices & Respective SQL Queries

5.) Order Cancellation Rate -> % of placed orders that end in cancellation (restaurant, rider, or customer).

Formula: cancelled_orders ÷ placed_orders × 100.

```
WITH placed AS (
  SELECT COUNT(*) AS placed_orders
  FROM fact_order_header oh
  JOIN dim_date d ON oh.date_key_ordered = d.date_key
  WHERE d.cal_date BETWEEN :start_date AND :end_date
),
cancelled AS (
  SELECT COUNT(DISTINCT c.order_id_nk) AS cancelled_orders
  FROM fact_cancellation c
  JOIN dim_date d ON c.date_key_cancel = d.date_key
  WHERE d.cal_date BETWEEN :start_date AND :end_date
)
SELECT
  placed_orders,
  cancelled_orders,
  ROUND(cancelled_orders * 100.0 / NULLIF(placed_orders,0), 2) AS cancellation_rate_percent
FROM placed CROSS JOIN cancelled;
```

Swiggy Data Warehouse Design

Grow Data Skills

Business Metrices & Respective SQL Queries

6.) Refund Rate -> % of delivered or cancelled orders that have any refund (full/partial).

Formula: refunded_orders ÷ placed_orders × 100.

```
WITH placed AS (
  SELECT COUNT(DISTINCT oh.order_id_nk) AS placed_orders
  FROM fact_order_header oh
  JOIN dim_date d ON oh.date_key_ordered = d.date_key
  WHERE d.cal_date BETWEEN :start_date AND :end_date
),
ref AS (
  SELECT COUNT(DISTINCT rf.order_id_nk) AS refunded_orders
  FROM fact_refund rf
  JOIN dim_date d ON rf.date_key_refund = d.date_key
  WHERE d.cal_date BETWEEN :start_date AND :end_date
)
SELECT
  placed_orders,
  refunded_orders,
  ROUND(refunded_orders * 100.0 / NULLIF(placed_orders,0), 2) AS refund_rate_percent
FROM placed CROSS JOIN ref;
```

Swiggy Data Warehouse Design

Grow Data Skills

Business Metrices & Respective SQL Queries

7.) Repeat Customer Rate (90-day) -> % of customers with ≥2 completed orders in last 90 days.

Formula: customers(orders≥2) ÷ customers(with ≥1 order) × 100.

```
WITH cust_orders AS (
  SELECT customer_sk, COUNT(DISTINCT order_id_nk) AS orders_90d
  FROM fact_order_header oh
  JOIN dim_date d ON oh.date_key_ordered = d.date_key
  WHERE d.cal_date BETWEEN date_sub(current_date(), 90) AND current_date()
    AND oh.status = 'DELIVERED'
  GROUP BY customer_sk
)
SELECT
  COUNT(*) AS active_customers_90d,
  SUM(CASE WHEN orders_90d >= 2 THEN 1 ELSE 0 END) AS repeat_customers_90d,
  ROUND(SUM(CASE WHEN orders_90d >= 2 THEN 1 ELSE 0 END) * 100.0 / NULLIF(COUNT(*),0), 2)
      AS repeat_customer_rate_percent
FROM cust_orders;
```

Swiggy Data Warehouse Design

Grow Data Skills

Business Metrices & Respective SQL Queries

8.) Restaurant Acceptance Rate -> % of order offers accepted by restaurants.

Formula: accepted_offers ÷ total_offers × 100.

```
SELECT
  r.restaurant_id_nk,
  r.brand,
  COUNT(*) AS total_offers,
  SUM(CASE WHEN a.accepted_flag = TRUE THEN 1 ELSE 0 END) AS accepted_offers,
  ROUND(SUM(CASE WHEN a.accepted_flag THEN 1 ELSE 0 END) * 100.0
        / NULLIF(COUNT(*),0), 2) AS acceptance_rate_percent
FROM fact_assignment a
JOIN dim_restaurant r ON a.restaurant_sk = r.restaurant_sk AND r.is_current = TRUE
JOIN dim_date d ON a.date_key = d.date_key
WHERE d.cal_date BETWEEN :start_date AND :end_date
GROUP BY r.restaurant_id_nk, r.brand
ORDER BY acceptance_rate_percent DESC;
```

Swiggy Data Warehouse Design

Grow Data Skills

Business Metrices & Respective SQL Queries

9.) Rider Utilization -> Share of rider shift time spent on “active” trip states (to-pickup, at-pickup, to-dropoff).

Formula: active_duration ÷ total_shift_duration × 100.

```
SELECT
  COUNT(*) AS shifts,
  SUM(rs.shift_duration_sec) AS total_shift_sec,
  SUM(rs.active_duration_sec) AS total_active_sec,
  ROUND(SUM(rs.active_duration_sec) * 100.0 / NULLIF(SUM(rs.shift_duration_sec),0), 2)
     AS rider_utilization_percent
FROM fact_rider_shift rs
JOIN dim_date d ON rs.date_key = d.date_key
WHERE d.cal_date BETWEEN :start_date AND :end_date;
```

Swiggy Data Warehouse Design

Grow Data Skills

Business Metrices & Respective SQL Queries

10.) Order Preparation SLA Hit Rate -> % of orders where restaurant ready_ts ≤ promised_ready_ts.

Formula: prep_sla_hits ÷ accepted_orders × 100.

```
SELECT
  COUNT(*) AS accepted_orders,
  SUM(CASE WHEN oh.ready_ts IS NOT NULL
            AND oh.ready_ts <= oh.promised_ready_ts
           THEN 1 ELSE 0 END) AS prep_sla_hits,
  ROUND(SUM(CASE WHEN oh.ready_ts IS NOT NULL
                  AND oh.ready_ts <= oh.promised_ready_ts
                 THEN 1 ELSE 0 END) * 100.0
        / NULLIF(COUNT(*),0), 2) AS prep_sla_hit_percent
FROM fact_order_header oh
JOIN dim_date d ON oh.date_key_ordered = d.date_key
WHERE d.cal_date BETWEEN :start_date AND :end_date
  AND oh.status IN ('ACCEPTED','DELIVERED');
```